SpaceX Falcon 9 Launch Analysis

IBM Applied Data Science Capstone Project

The purpose of this project was to analyse SpaceX Falcon 9 launch data and investigate factors that may affect successful rocket landings.

Data was collected, cleaned and analysed using Python and SQL. Charts and an interactive dashboard were used to explore launch sites, payload masses and booster versions.

The analysis showed that most launches in the prepared dataset were successful. The results helped identify patterns in SpaceX launch performance.

SpaceX is a company that develops and launches rockets and spacecraft.

The Falcon 9 rocket is designed to carry payloads into space. One important feature is that its first-stage booster can return to Earth and land for reuse.

The aim of this project is to analyse historical Falcon 9 launch data and understand how factors such as launch site, payload mass and booster version relate to launch success.

The project used historical SpaceX Falcon 9 launch data.

The data was collected using the SpaceX API and web scraping.

The dataset included information such as:

- Launch dates
- Launch sites
- Payload masses
- Booster versions
- Launch outcomes

This information was used to investigate patterns and understand factors related to launch success.

The collected data needed to be cleaned before analysis.

Python and Pandas were used to:

- Remove unnecessary information.
- Handle missing values.
- Convert data into the correct formats.
- Organise launch dates and other details.
- Prepare the dataset for analysis.

This process helped make the data more accurate and suitable for analysis.

Exploratory Data Analysis was used to understand the SpaceX launch data.

Python libraries such as Pandas, Matplotlib and Seaborn were used to explore the data and create visualisations.

The analysis focused on:

- Launch success and failure.
- Launch sites.
- Payload mass.
- Booster versions.

Charts helped identify patterns and compare launch performance.

SQL was used to explore the SpaceX launch dataset and answer important questions.

The analysis included:

- Counting launches at each launch site.
- Comparing successful and unsuccessful launches.
- Finding the launch site with the most launches.
- Examining payload mass and launch outcomes.

SQL queries helped organise the data and identify useful results.

Folium was used to create an interactive map showing the locations of SpaceX launch sites.

The map helped show where the launches took place and allowed the launch sites to be viewed geographically.

This visualisation made it easier to understand the geographical distribution of SpaceX launch activities.

A dashboard was developed using Python, Plotly and Dash.

It included:

- A dropdown menu to select a launch site.
- A pie chart showing launch outcomes.
- A slider to select a payload mass range.
- A scatter plot comparing payload mass and landing outcomes.

The dashboard allowed users to explore launch data interactively and compare results.

Machine learning was used to investigate whether launch success could be predicted from the available data.

Different classification algorithms were explored and compared.

The models were evaluated using performance measures such as accuracy and a confusion matrix.

The aim was to identify which model performed best when predicting launch outcomes.

The analysis of the prepared dataset showed that:

- CCAFS SLC-40 had the largest number of launches, with 34.
- Three launch sites had a 100% success rate.
- CCAFS LC-40 had one unsuccessful launch.
- The F9 v1.1 booster category had a lower success rate than the other recorded F9 categories.

These findings show how launch sites and booster versions can be compared using historical data.

This project showed how data science can be used to analyse SpaceX Falcon 9 launch performance.

Python, SQL, visualisations and an interactive dashboard helped explore the launch data and identify patterns.

The analysis showed that most recorded launches were successful. However, the findings should be interpreted carefully because the dataset contains a limited number of unsuccessful launches.

Future analysis could use more launch records and additional information to improve the understanding and prediction of launch outcomes.

This project brought together different data science tools to investigate SpaceX launch performance.

The use of Python, SQL, visualisations and an interactive dashboard made it easier to explore and understand the data.

Future improvements could include:

- Adding more launch records.
- Using more information about each launch.
- Improving the predictive models.
- Making the dashboard easier to explore.